

Internship in Physical Organic Chemistry
Structure Characterisation Sheet — Negative Ion Mode

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Fill in this sheet after completing the positive mode sheet. Sections 1–3 mirror the positive mode sheet. Section 4 is the cross-comparison: answer it only after both sheets are complete.

Section 1 — MS1 spectrum (negative ion mode)

Compound: _____ **Date:** _____

Solvent: _____ **Concentration:** _____ **Raw file:** _____

List all peaks observed in the negative MS1 spectrum.

Observed m/z	Proposed identity	Status	Notes
_____	_____	☐ confirmed tentative	☐ _____
_____	_____	☐ confirmed tentative	☐ _____
_____	_____	☐ confirmed tentative	☐ _____
_____	_____	☐ confirmed tentative	☐ _____
_____	_____	☐ confirmed tentative	☐ _____
_____	_____	☐ confirmed tentative	☐ _____

Isotope simulation check (molecular ion):

Observed m/z : _____ Theoretical m/z : _____ Mass error (ppm): _____

Isotope pattern matches simulation: ☐ Yes ☐ No ☐ Partially

Precursor selected for MS/MS: _____ **Reason:** _____

Section 2 — MS/MS fragment table (negative ion mode)

Precursor ion (m/z): _____ NCE used for this table: _____ Raw file: _____

Fragment m/z	Neutral loss (Da)	Proposed formula of neutral	Proposed assignment / notes
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Effect of NCE on fragmentation pattern:

Section 3 — RDA argument and structure sketches (negative ion mode)

RDA fragment observed at m/z : _____ Neutral loss from precursor:
_____ Da

Proposed identity of charged fragment: _____

Proposed identity of neutral fragment (lost): _____

Structure sketch: (draw here)

Is this assignment consistent with the Anchor 2 paper?

☐ Yes ☐ No ☐ Partially ☐ Not yet checked

Notes: _____

Section 4 — Cross-comparison of positive and negative ion mode

Complete this section only after both the positive and negative mode sheets are fully filled in. Answer each question in complete sentences. Your answers feed directly into scratch section 2W-e.

1. Which fragment ions appear in both modes? Which appear in only one mode?

2. Does the RDA cleavage produce the same neutral loss in both modes, or different losses?

What does this tell you about which fragment retains the charge in each mode?

3. Is the fragmentation pattern at medium NCE richer in positive or negative mode?

What might explain this?

4. Based on your observations, which ionisation mode gives the most informative data

for the RDA reaction you are studying? Justify your answer in 2–3 sentences.
